

- 2 (a) Fig. 2.1 is a transmission electron micrograph of a transverse section through a blood capillary that is surrounded by actively respiring cells.

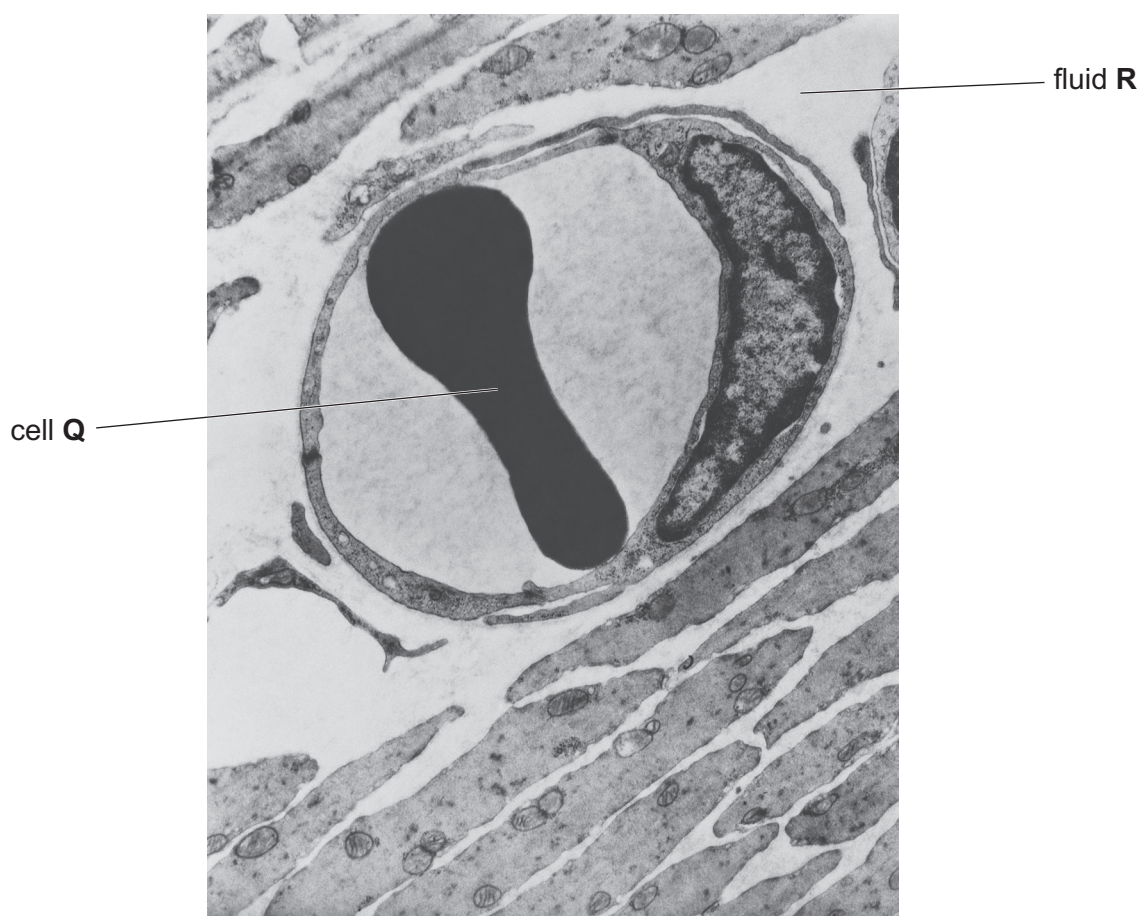


Fig. 2.1

- (i) Identify cell **Q** shown in Fig. 2.1.

..... [1]

- (ii) Fluid **R** is formed from blood in the blood capillary.

State **and** explain **one** difference between the composition of fluid **R** and the composition of the blood in the capillary.

difference

explanation

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..... [3]

- (b) (i) The cell surface membranes of the respiring cells have a role in regulating the exchange of substances between the cells of the body and the blood.

Draw a diagram in the space below to show the arrangement of phospholipid molecules in the cell surface membrane.

[2]

- (ii) Cholesterol is a molecule found in the cell surface membrane.

Describe **one** role of cholesterol in the cell surface membrane.

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..... [1]

- (c) Nicotine is one of the components of tobacco smoke. Nicotine can cross cell surface membranes and enter the bloodstream.

- (i) Describe the short-term effects of nicotine on the cardiovascular system.

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..... [2]

- (ii) Nicotine is absorbed from the blood plasma into the cells of the capillary wall. Scientists have suggested that nicotine is transported from the blood plasma into these cells using a type of co-transporter mechanism.

Fig. 2.2 shows how nicotine may be transported from the blood plasma into a cell.

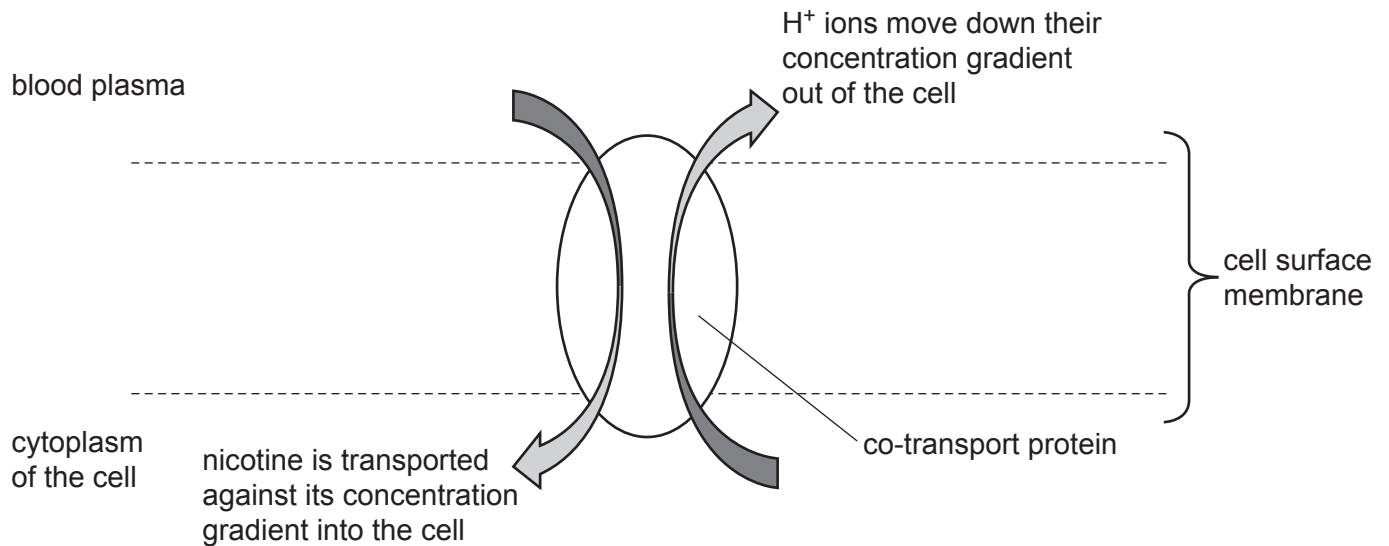


Fig. 2.2

The co-transporter mechanism shown in Fig. 2.2 is different from the co-transporter mechanism that moves sucrose into the cytoplasm of a companion cell in phloem tissue.

Outline the similarities **and** differences between the co-transport of nicotine and the co-transport of sucrose.

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..... [3]

[Total: 12]

Question 3 starts on page 8.